Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Applied AI Engineer with 8+ years of production Python engineering and hands-on agentic AI development. Built **AgentProbe** -- a 3-agent LangGraph production system (Orchestrator, RAG Tracer, Auto Fix) with a 5-metric LLM-as-judge eval engine, ChromaDB RAG pipeline, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) -- the exact architecture Cohere's Applied AI Engineers build for enterprise customers. Shipped 20+ production modules at scale (1M+ records, 100K+ MAU) at Entrata; MS Data Analytics Engineering, GMU (GPA 3.87, May 2026). Demo: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

LangGraph Multi-Agent Systems | LLM Evaluation & LLM-as-Judge | RAG Pipelines & ChromaDB | Multi-Provider LLM Routing | Agent Observability & Tracing | FastAPI Production APIs | Python (8+ yrs production) | Prompt Engineering | Enterprise-Scale Reliability | ReAct & Agentic Patterns | CI/CD & Quality Gates | Docker & Microservices

EXPERIENCE

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Virginia, USA

- Delivered targeted Python and database mentoring for 30+ undergraduates -- strengthened data structures, algorithms, and debugging skills across iterative sprint-style delivery cycles.
- Graded assignments with rubric-aligned feedback and developed supplementary Python exercises, accelerating concept reinforcement and course feedback loops.

George Mason University

Aug 2024 - Aug 2025

Research Associate - Virginia, USA

- Built regression models (R^2 = 0.977) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries -- multi-source data collection, transformation, validation, and stakeholder reporting.
- Applied Isolation Forest anomaly detection to automate data integrity checks across 3 research pipelines -- surfaced anomalies that manual review missed, improving downstream model reliability.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership -- improved research comprehension across two cohorts by 40%.

Entrata India (Db Xento Systems)

Oct 2018 - Aug 2024

Software Engineer - Pune, India

- Applied Python, PyTorch, and OpenAI API to automate data transformation pipelines across 1.6M financial records -- production AI on regulated enterprise data with output validation and quality gates.
- Reduced deployment times by 40% by building 20+ full-stack microservices (Python, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints for 100K+ monthly active users -- 99.5% uptime sustained over 6+ consecutive months.
- Cut microservices execution time by 50% by leading Docker-based containerization for 30%+ of modules; improved production query performance by 30% by tuning 200+ SQL queries across all client environments.
- Achieved 100% PHI compliance by migrating all client data to AWS S3 with Libsodium-encrypted pipelines -- zero security incidents across all major releases.
- Conducted 300+ peer code reviews and mentored 5+ junior engineers, enforcing secure coding practices and TDD/BDD standards across all major releases.

Entrata India

Jan 2017 - Sep 2018

Associate Software Engineer / Trainee - Pune, India

- Improved system stability by 35% by developing 20+ full-stack property management modules (Python, PostgreSQL) -- 99.5% uptime and 25% faster report generation.
- Resolved 100+ high-priority bugs; delivered production-ready APIs from day one using Git-based workflows, secure coding standards, and TDD practices.

PROJECTS

AgentProbe: Agentic AI Testing Toolkit IN PROGRESS

Python, LangGraph, LangChain, FastAPI, ChromaDB, Anthropic, OpenAI, Groq, Ollama, React, Docker, GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and **multi-provider model routing** (Anthropic, OpenAI, Groq, Ollama) -- deployed as a 15-endpoint FastAPI backend running 100% locally at zero cost. Automated agent quality gates with a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with CI/CD integration -- failing runs blocked automatically before production. Delivered end-to-end observability via React dashboard with trace-level agent monitoring -- ChromaDB RAG pipeline tracking per-run metrics, regression drift, retrieval quality, and latency across concurrent agent test suites.

AI on FHIR: NLP-Driven Medical Records Query System

Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Reduced manual clinical retrieval time by ~60% by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings -- HIPAA-compliant RBAC across structured medical records. Demonstrates LLM application on regulated enterprise data.

CrashClarity: Large-Scale Road Accident Analysis (104K Records)

Python, R, scikit-learn, Random Forest, Logistic Regression, ggplot2, Feature Engineering, GitHub Pages

Achieved 85%+ accuracy on 104K+ UK accident records by building a Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables -- deployed interactive dashboard on GitHub Pages for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia 2024-2026

GPA: 3.87

B.Tech Computer Science & Engineering — Sree Venkateshwara College of Engineering, India 2013-2017

TECHNICAL SKILLS

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, Tool Calls, LLM Router, ChromaDB, RAG Pipelines, Embeddings, LLM Evaluation, LLM-as-Judge, Agent Observability, Agent Tracing, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama

Languages: Python, JavaScript, TypeScript, SQL, PHP, R, Bash, HTML/CSS

Backend & APIs: FastAPI, Flask, RabbitMQ, Microservices, GraphQL, PostgreSQL, MySQL, Oracle, NoSQL

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions)

ML: PyTorch, TensorFlow, XGBoost, scikit-learn, Isolation Forest, Random Forest, BERT, Hugging Face, spaCy

Testing: TDD/BDD, Selenium, Integration Testing, Secure Coding